Writing Equations in Algebraic Form

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Equation

An **equation** is a mathematical statement that shows that two expressions are equal using the equal sign ($=$).

$$x + 3 = 10$$

Left side $=$ Right side

Algebraic Equation

An **algebraic equation** contains one or more variables, numbers, and mathematical operations.

$$2x + 5 = 17$$

Variable: x
Number: 5, 17, 2
Operations: $\times$ and $+$

Verbal Statement

A **verbal statement** describes a mathematical relationship using words.

We can translate it into an algebraic equation.

"A number increased by 4 is 10."

$$x + 4 = 10$$

Steps in Writing an Equation

Step 1

Identify the **unknown quantity** and represent it with a variable.

"A number" $\rightarrow x$

Step 2

Identify the **mathematical operation** or relationship described.

"increased by" $\rightarrow$ addition

Step 3

Translate the words into mathematical symbols.

"increased by 4" $\rightarrow x + 4$

Step 4

Use the **equal sign** to show the relationship between the two quantities.

"is 10" $\rightarrow = 10$

$$x + 4 = 10$$

Remember the 4 Steps

Unknown $\rightarrow$ variable

Operation $\rightarrow$ identify

Symbols $\rightarrow$ translate

Equal sign $\rightarrow$ connect

Mnemonic: **U O S E** — "**U**se **O**ur **S**martest **E**quations!"

Common Verbal Phrases

Addition

Sum, added to, increased by, plus

$+$

"The sum of a number and 5"

$x + 5$

Subtraction

Difference, subtracted from, decreased by, less than

$-$

"A number decreased by 3"

$$x - 3$$

Multiplication

Product, times, multiplied by

$\times$

"Three times a number"

$3x$

Division

Quotient, divided by

$$\div$$

"The quotient of a number and 5"

$$\frac{x}{5}$$

Equality

Is, equals, is equal to

$$=$$

"A number is 12."

$$x = 12$$

Special Phrases

Twice a number: $2x$

Three times a number: $3x$

A number squared: x^2

Watch Out: "Less Than"

"5 less than a number" means start with the number, then subtract 5.

$$x - 5$$

Not $5 - x$!

Examples

Example 1

A number increased by 8 is equal to 20.

Unknown: x

Operation: addition

$$x + 8 = 20$$

Example 2

Three times a number is 21.

Unknown: x

Operation: multiplication

$$3x = 21$$

Example 3

The quotient of a number and 4 is 6.

Unknown: x

Operation: division

$$\frac{x}{4} = 6$$

Example 4

Twice a number plus 5 equals 17.

"Twice a number" $\rightarrow 2x$

"plus 5" $\rightarrow +5$

$$2x + 5 = 17$$

Example 5

Three times a number decreased by 2 is 13.

"Three times a number" $\rightarrow 3x$

"decreased by 2" $\rightarrow -2$

$$3x - 2 = 13$$

Practice Exercises

Directions

Write each verbal statement as an algebraic equation.

Use x to represent the unknown number unless another variable is specified.

Practice 1

A number increased by 9 is equal to 25.

Practice 2

The difference between a number and 7 is 18.

Practice 3

Four times a number is 36.

Practice 4

The quotient of a number and 5 is 8.

Practice 5

Twice a number plus 6 equals 30.

Practice 6

Three times a number decreased by 4 is 20.

Practice 7

The sum of a number and 12 is twice the number.

Practice 8

Five less than a number is equal to 14.

Practice 9

The sum of twice a number and 7 is 31.

Practice 10

The quotient of a number decreased by 3 and 4 is 6.

Activities

Directions

Translate each verbal statement into an algebraic equation.

Activity 1

A number increased by 11 is equal to 29.

Activity 2

The difference between a number and 9 is 16.

Activity 3

Six times a number is 42.

Activity 4

The quotient of a number and 7 is 5.

Activity 5

Twice a number plus 8 equals 34.

Activity 6

Four times a number decreased by 5 is 27.

Activity 7

The sum of a number and 15 is twice the number.

Activity 8

Six less than a number is equal to 19.

Activity 9

The sum of twice a number and 9 is 35.

Activity 10

The quotient of a number decreased by 5 and 3 is 8.

Thank you!